

Sea and water surface temperature

Introduction

Infrared data are widely used to estimate the surface temperature of the land and waterbodies. This activity will compare the sea surface temperature from a few different sensors in a region of interest. A set of files are given for a region off the west coast of Canada from which SST can be derived, with some suggested follow up activities.

Some reading is given below related to this topic:

Merchant CJ et al., Satellite-based time-series of sea-surface temperature since 1981 for climate applications. Sci Data. 2019 Oct 22;6(1):223. doi: 10.1038/s41597-019-0236-x

Minnett, P.J. et al., 2019, Half a century of satellite remote sensing of sea-surface temperature, Remote Sensing of Environment, 233, 111366.

Jiménez-Muñoz J.C., Sobrino J.A., Skoković D., Mattar C., Cristóbal J.
Land surface temperature retrieval methods from Landsat-8 thermal infrared sensor data
IEEE Geosci. Remote Sens. Lett., 11 (2014), pp. 1840-1843

Part 1

Data files are provided from Landsat-8 with a short test script to plot the surface temperature from inverting Planck's equation and assuming an emissivity of 1.0. Files are also provided from Sentinel-3 where the sea surface temperature (SST) has already been retrieved with a short test script to plot the SST.

1. Run each test script to plot the surface temperature. Comment on the similarities and differences you see between the plots. These can be due to various factors such as spatial resolution, acquisition time, atmospheric conditions and sensor artifacts.
2. Using equation 5 and Table 1 from Jimenez-Munoz et al. obtain an estimate of the sea surface temperature of from Landsat. Note that although the equation given is for land surface temperature it has also been used in several studies for waterbodies. In this case, because the emissivity is close to 1.0 in the thermal infrared bands you can simplify equation 5 to neglect the $\Delta\epsilon$ and $(1 - \epsilon)$ terms. Compare this with the brightness temperature from inverting Planck's equation and comment on the differences

Part 2

For this part you can do either (a) or (b) or make a different suggestion, depending on your interest

- a) Obtain Landsat and Sentinel-3 or VIIRS data for a different region and repeat the surface temperature retrieval for a chosen time period. Comment on the differences between the temperature retrievals and which one is more suitable for a given application.
- b) For the west coast region from Part 1 expand the dataset to cover more dates and compare the retrieved SSTs with glider data from <https://cproof.uvic.ca/>. You may wish to use VIIRS data instead of Sentinel-3 (they both have capability to measure SST). Discuss the differences between the temperature measured from the glider (at the appropriate level, close to the surface) and the SST.

Some links you may find useful:

<https://ladsweb.modaps.eosdis.nasa.gov/search/>

<https://browser.dataspace.copernicus.eu/>

<https://earthexplorer.usgs.gov/>